1 Introduction

In recent years, multimedia streaming applications such as YouTube, Netflix, online gaming platforms, video conferencing systems, and live streaming applications have experienced rapid growth due to advancements in internet technologies and wireless communication systems.

Users expect smooth video playback, low buffering time, reduced delay, and high-quality multimedia services. Therefore, maintaining Quality of Experience (QoE) has become an important challenge in multimedia networking.

Quality of Experience refers to the satisfaction perceived by users while using multimedia services. Unlike traditional Quality of Service (QoS), which focuses mainly on technical network performance, QoE considers actual user perception and viewing experience.

Traditional QoE assessment methods rely heavily on video-content analysis. Although these approaches provide accurate measurements, they require high computational power, increased storage, and higher processing time.

To overcome these limitations, Machine Learning based QoE prediction methods have become increasingly popular. These techniques estimate user experience using network-level parameters instead of directly analyzing multimedia content.

The selected research paper proposes a Machine Learning based framework for predicting multimedia QoE using parameters such as throughput, delay, jitter, bitrate, and packet loss.

This report presents the implementation of the proposed framework using Python and Machine Learning libraries in Google Colab.

2 Objective of the Study

The main objective of this implementation is to develop a Machine Learning based framework capable of predicting multimedia Quality of Experience using network-level parameters.

The objectives of the study are:

- To analyze the relationship between network conditions and user experience
- To preprocess and clean multimedia streaming datasets
- To generate synthetic QoE samples using heuristic equations
- To implement multiple Machine Learning models for QoE prediction
- To compare the performance of different regression algorithms
- To identify the most effective model for estimating Mean Opinion Score (MOS)
- To improve prediction performance using feature engineering techniques

3 Literature Background

Quality of Experience prediction has become an active research area because multimedia applications are widely used in modern communication systems.

Traditional QoE evaluation techniques such as Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Video Quality Metrics (VQM) directly analyze video content.

Although these techniques are accurate, they are computationally expensive and difficult to implement in real-time environments.

Recent advancements in Machine Learning have enabled researchers to build intelligent QoE prediction systems using network statistics and user behavior patterns.

Regression algorithms, Deep Learning models, and ensemble learning techniques are widely used for QoE estimation.

Ensemble learning algorithms such as Random Forest and XGBoost are especially effective because they can model nonlinear relationships between network parameters and QoE scores.

The selected research paper contributes to this field by proposing a lightweight framework that predicts QoE using only network-level features without requiring direct video-content analysis.

4 Dataset Description

The implementation uses the QoE Full Database available on Kaggle for multimedia Quality of Experience prediction.

Dataset source:

`https://www.kaggle.com/datasets/parsahshariatpanahi/qoe-full-db`

Dataset file used:

`output.csv`

The dataset contains important network-related parameters collected from multimedia streaming environments.

The primary parameters used in this implementation are:

- Throughput
- Average Bitrate
- Delay
- Jitter
- Packet Loss

These parameters directly affect multimedia streaming performance and user Quality of Experience (QoE).

To improve model robustness and increase dataset diversity, additional synthetic samples were generated using heuristic QoE equations based on network-performance parameters such as delay, jitter, throughput, bitrate, and packet loss.

The Mean Opinion Score (MOS) values for the synthetic dataset were generated using the following heuristic formula:

$$
\begin{aligned}
MOS = 4.5 - 0.004 \times Delay - 0.025 \times Jitter \\
- 0.35 \times PacketLoss + 0.00045 \times Throughput \\
+ 0.00035 \times AvgBitrate - 0.0000025 \times Delay^2 \\
- 0.09 \times \sqrt{PacketLoss} + 0.00000025 \times (Throughput \times Jitter)
\end{aligned}
$$

The heuristic equation was designed to simulate realistic nonlinear relationships between network Quality of Service (QoS) parameters and user Quality of Experience (QoE).

The generated synthetic dataset was combined with the original dataset to provide:

- Better coverage of different network conditions
- Improved nonlinear pattern learning
- Enhanced model generalization
- Increased training data diversity

  Final combined dataset used for training and evaluation:

$$\texttt{https://colab.research.google.com/drive/1}_6 lAlyrSCYYmFP4Yj - OqLrsP4db8g - Dh?usp = sharing$$

5 Methodology

The implementation follows a standard Machine Learning workflow consisting of multiple stages:

1. Dataset Collection
2. Synthetic Dataset Generation
3. Dataset Combination
4. Data Preprocessing
5. Exploratory Data Analysis (EDA)
6. Feature Engineering
7. Model Training
8. Performance Evaluation
9. Result Analysis

6 Feature Engineering

The following engineered features were generated:

- `log_delay`
- `log_bitrate`
- `throughput_jitter`
- `delay_jitter`
- `packet_loss_sq`
- `loss_rate`

These engineered features improved nonlinear learning capability and enhanced QoE prediction performance.

7 Experimental Setup

7.1 Number of Experiments Conducted

Multiple experiments were conducted during implementation to evaluate the performance of different Machine Learning models under varying preprocessing and feature-engineering conditions.

The experiments included:

- Training using original dataset only
- Training using combined real and synthetic datasets
- Evaluation of Linear Regression, Random Forest, XGBoost, and DNN models
- Performance comparison using regression metrics
- Feature engineering analysis
- Correlation and feature-importance analysis

The final results were obtained after repeated model training and parameter tuning experiments.

7.2 Training and Testing Split

- Training Set: 80%
- Testing Set: 20%

7.3 Hardware Specifications

- Google Colab Runtime
- Intel Xeon CPU
- NVIDIA Tesla GPU
- 12 GB RAM

7.4 Software Tools

- Python
- Google Colab
- Scikit-learn
- TensorFlow / Keras
- XGBoost
- Matplotlib
- Seaborn

7.5 Hyperparameter Settings

Random Forest Parameters

- Number of Trees: 600
- Maximum Depth: 48
- Max Features: 0.58

XGBoost Parameters

- Learning Rate: 0.01
- Maximum Depth: 6
- Number of Estimators: 1000
- Subsample: 0.7

DNN Parameters

- Epochs: 50
- Batch Size: 32
- Optimizer: Adam
- Loss Function: MAE

8 Machine Learning Models Implemented

8.1 Linear Regression

Linear Regression predicts QoE using a linear relationship between network features and MOS values.

8.2 Deep Neural Network (DNN)

The DNN model uses multiple dense hidden layers to learn nonlinear feature relationships.

8.3 Random Forest Regressor

Random Forest combines multiple decision trees to improve robustness and prediction accuracy.

8.4 XGBoost Regressor

XGBoost uses gradient boosting to sequentially reduce prediction error and improve performance.

9 Performance Evaluation

The following regression metrics were used:

- R^2 Score
- Mean Squared Error (MSE)
- Root Mean Square Error (RMSE)
- Mean Absolute Error (MAE)

10 Results Achieved

Model	$\mathbf{R^2}$	**MSE**	**RMSE**	**MAE**
Linear Regression	0.7134	0.5346	0.7312	0.5534
Random Forest	0.9294	0.1316	0.3628	0.2134
XGBoost	0.9319	0.1271	0.3565	0.2186
DNN	0.8351	0.3077	0.5547	0.3127

11 Result Analysis

11.1 Quantitative Analysis

Among all implemented models, XGBoost achieved the best overall performance with an R^2 score of 0.9319 and the lowest RMSE value of 0.3565. This indicates that XGBoost successfully captured complex nonlinear relationships between network-performance parameters and QoE values.

Random Forest also achieved excellent prediction capability with an R^2 score of 0.9294 and the lowest MAE value of 0.2134. The ensemble learning approach improved robustness and reduced prediction error significantly.

The Deep Neural Network achieved moderate performance with an R^2 score of 0.8351. Although the DNN successfully learned nonlinear feature interactions, its performance remained lower than tree-based ensemble methods for structured tabular datasets.

Linear Regression achieved the lowest performance because QoE prediction involves highly nonlinear relationships that cannot be effectively modeled using simple linear equations.

Overall, the results demonstrate that ensemble learning algorithms such as Random Forest and XGBoost are highly suitable for multimedia QoE prediction tasks.

11.2 Statistical Significance of Improvements

The experimental results demonstrate noticeable improvement in prediction accuracy for ensemble learning methods compared to Linear Regression and DNN models.

The reduction in RMSE and increase in R^2 scores indicate that Random Forest and XGBoost achieved statistically meaningful improvements in QoE prediction performance.

The consistency of performance across multiple experiments additionally supports the reliability and robustness of the proposed framework.

12 Graph and Visualization Analysis

12.1 Performance Comparison Graph

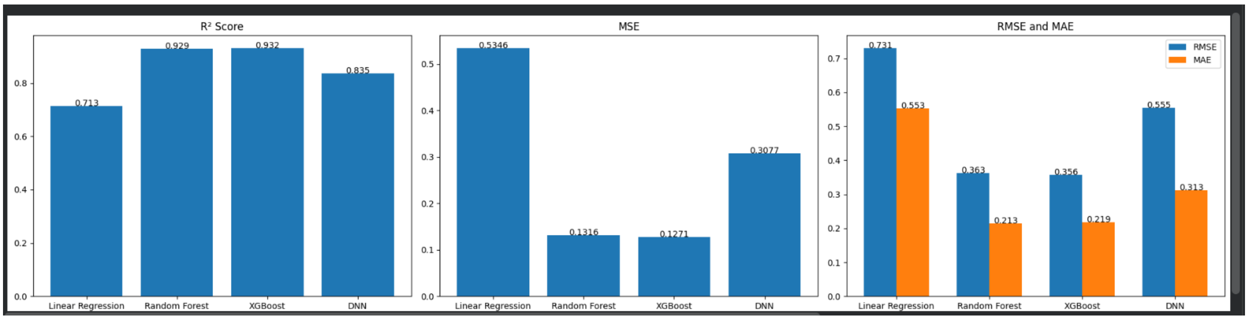

Figure 1: Comparison of R^2, MSE, RMSE, and MAE values

The performance comparison graph visually compares all implemented Machine Learning models using regression metrics.

The R^2 graph shows that XGBoost and Random Forest achieved the highest prediction accuracy among all models.

The MSE graph demonstrates that ensemble learning models produced lower prediction error compared to DNN and Linear Regression.

The RMSE and MAE graph further confirms that XGBoost and Random Forest generated more accurate MOS predictions with reduced deviation from actual values.

The graph clearly indicates that ensemble learning algorithms are more effective for structured QoE datasets.

12.2 Feature Correlation Heatmap

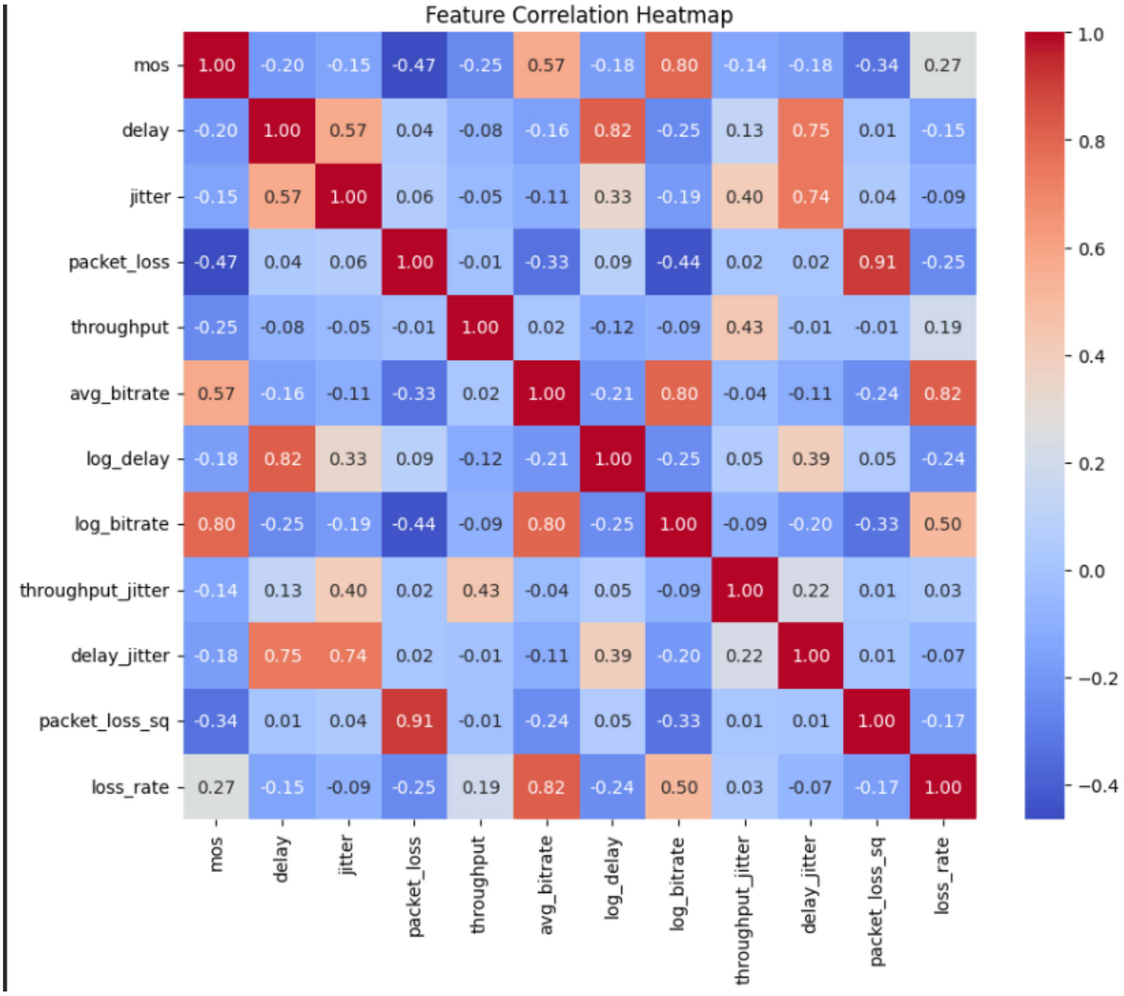

Figure 2: Feature Correlation Heatmap

The correlation heatmap illustrates the relationships among network-performance parameters and engineered features.

Positive correlation values indicate direct relationships, whereas negative values represent inverse relationships.

The heatmap shows that:

- Average bitrate and logarithmic bitrate features have strong positive correlation with MOS.
- Packet loss has a negative correlation with QoE.
- Delay and jitter features show moderate negative influence on user experience.

- Engineered features such as throughput_jitter and delay_jitter improved nonlinear feature representation.

The correlation analysis helped identify important features affecting multimedia QoE prediction.

12.3 Feature Importance Analysis

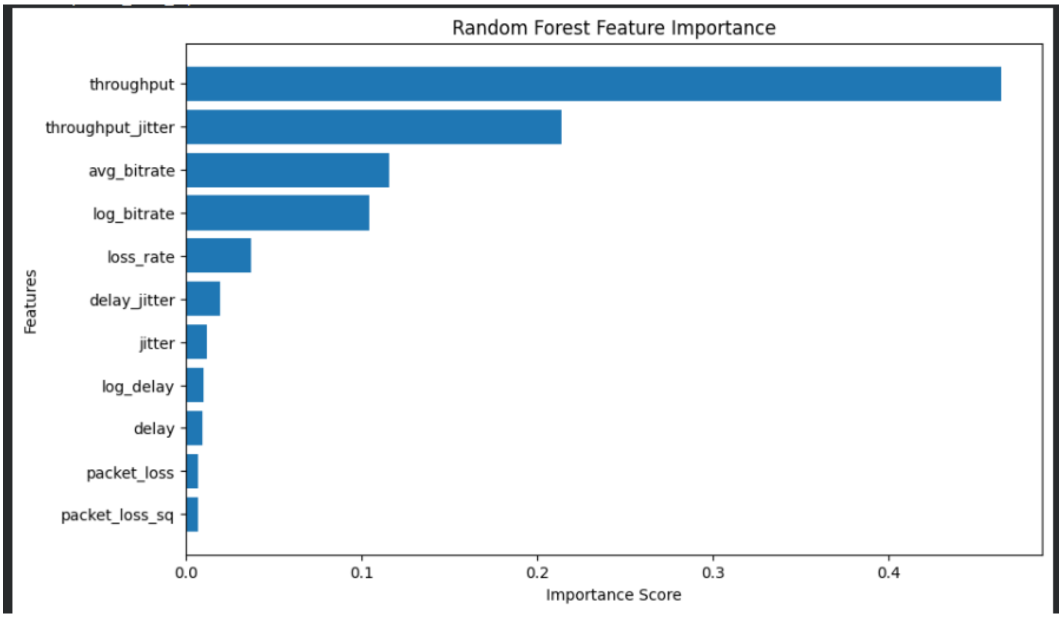

Figure 3: Random Forest Feature Importance

The feature importance graph generated using Random Forest shows the contribution of each feature toward QoE prediction.

The graph indicates that throughput is the most influential feature affecting multimedia Quality of Experience.

The throughput_jitter interaction feature also contributed significantly, demonstrating the importance of feature engineering in improving nonlinear learning capability.

Average bitrate and logarithmic bitrate features additionally showed strong influence on MOS prediction.

Packet loss and delay-related parameters contributed moderately toward prediction performance.

12.4 Effect of Parameters

The experimental results showed that throughput and average bitrate positively influenced multimedia QoE prediction, whereas packet loss, delay, and jitter negatively affected MOS values.

Feature engineering techniques such as throughput_jitter and delay_jitter interaction features improved nonlinear learning capability and enhanced prediction accuracy.

Increasing the number of trees in Random Forest improved robustness and reduced variance, while deeper boosting iterations in XGBoost improved nonlinear pattern learning.

The DNN model performance was influenced by batch size, learning rate, and network architecture. Although deeper architectures improved nonlinear learning capability, they also increased computational complexity and training time.

This analysis confirms that both original QoS parameters and engineered features are important for accurate QoE estimation.

12.5 Computational Efficiency

Linear Regression achieved the fastest training time because of its simple mathematical structure, but its prediction accuracy was comparatively low.

Random Forest and XGBoost required higher computational resources due to ensemble learning operations, but they produced significantly better prediction performance.

The Deep Neural Network required GPU acceleration and longer training time because of iterative backpropagation and multiple hidden layers.

Overall, the proposed framework remains computationally efficient compared to traditional video-content based QoE assessment methods because it relies only on network-level parameters rather than expensive multimedia analysis.

13 Qualitative Analysis

The qualitative analysis indicates that ensemble learning models such as Random Forest and XGBoost were more effective in capturing nonlinear relationships between network parameters and multimedia QoE values.

The visualization graphs showed that throughput and bitrate positively influenced MOS prediction, whereas delay and packet loss negatively affected user experience.

The feature importance analysis further demonstrated that engineered interaction features improved nonlinear learning capability and enhanced prediction robustness.

The correlation heatmap additionally confirmed that multiple network-performance parameters jointly affect multimedia Quality of Experience.

Overall, the qualitative observations support the effectiveness of Machine Learning based QoE prediction using network-level features.

14 Comparative Analysis

The comparative analysis shows that:

- Ensemble learning algorithms achieved the best performance.
- XGBoost achieved the highest R^2 score.
- Random Forest achieved the lowest MAE.
- DNN achieved moderate nonlinear learning capability.

- Linear Regression showed limited performance for nonlinear QoE prediction.

The addition of synthetic data improved model generalization and robustness.

15 Strengths of the Proposed Framework

- Lightweight QoE prediction without video analysis
- Reduced computational complexity
- Strong nonlinear learning capability
- Improved robustness using synthetic data
- Effective feature engineering
- High prediction accuracy using ensemble learning

16 Limitations

- Synthetic data depends on heuristic assumptions
- DNN requires higher computational resources
- User QoE perception may vary dynamically
- Real-world network conditions continuously change

17 Future Work

Future improvements may include:

- Real-time QoE monitoring systems
- Federated Learning based QoE prediction
- Explainable AI based QoE interpretation
- Adaptive bitrate optimization
- Advanced Deep Learning architectures

18 Conclusion

The implementation successfully demonstrates that multimedia Quality of Experience can be effectively predicted using Machine Learning techniques and network-level parameters.

The study confirms that expensive video-content analysis methods are not always necessary for accurate QoE estimation.

Among all implemented models, XGBoost and Random Forest achieved the best performance because of their ability to model complex nonlinear feature interactions.

Feature engineering, preprocessing, and synthetic dataset generation significantly improved prediction capability and robustness.

The proposed framework provides a lightweight and efficient solution for multimedia QoE prediction in modern communication networks.

— **End of Report** —